

ITSimplera Solutions



Cybersecurity Internship

Week 5 Task Report

Network Scanning & Vulnerability Assessment on Metasploitable 2

Submitted by

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Introduction

This report covers the work I completed in Week 5 of my internship. The main goal this week was to scan a target machine, find out which services were running on it, and check those services for known security weaknesses.

The target machine used for all the scans was Metasploitable 2, a virtual machine that is deliberately built with security flaws so it can be used for practice in a safe, controlled lab. I used Nmap to scan the machine, look at open ports, and identify running services. I then researched the vulnerabilities linked to those services and noted down ways to fix them.

This document is organized into six tasks. The first three tasks cover scanning the target and listing the services and vulnerabilities found. Task 4 compares different Nmap scan types. Task 5 looks at the security risks in more detail and how to fix them. Task 6 puts everything together into a full vulnerability assessment report.

Task 1 – Network Enumeration

I ran the following Nmap commands against the target IP address 10.0.2.4 to find open ports and gather information about the machine.

```
nmap 10.0.2.4
```

```
└─$ nmap 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 02:11 EDT
Nmap scan report for 10.0.2.4
Host is up (0.00012s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:88:8D:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

```
nmap -sV 10.0.2.4
```

```
└─$ nmap -sV 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 02:13 EDT
Nmap scan report for 10.0.2.4
Host is up (0.00016s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:88:8D:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux;
```

```
nmap -O 10.0.2.4
```

```
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 1.62 seconds
```

```
nmap -A 10.0.2.4
```

```
└─(kali㉿kali)-[~]
└─$ nmap -A 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 02:18 EDT
Nmap scan report for 10.0.2.4
Host is up (0.00088s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 10.0.2.3
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
```

```
nmap -p- 10.0.2.4
```

```

22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
38547/tcp open  unknown
43383/tcp open  unknown
54653/tcp open  unknown
57627/tcp open  unknown

```

Task 2 — Service Enumeration

The table below lists every open port found on the target, along with the service running on it and what that service is normally used for.

Port	Protocol	Service	Version	Status	Purpose
21	TCP	FTP (vsftpd)	vsftpd 2.3.4	Open	Uploading and downloading files.
22	TCP	SSH	OpenSSH 4.7p1	Open	Secure remote login and commands.
23	TCP	Telnet	Linux telnetd	Open	Remote command-line access (not encrypted).
25	TCP	SMTP	Postfix	Open	Sending and forwarding email.
53	TCP	DNS	ISC BIND 9.4.2	Open	Turning domain names into IP addresses.
80	TCP	HTTP	Apache 2.2.8	Open	Hosting and serving web pages.
111	TCP	RPCBind	Version 2	Open	Maps RPC services to network ports.
139	TCP	NetBIOS	Samba 3.x-4.x	Open	Windows file and printer sharing.
445	TCP	SMB	Samba 3.0.20	Open	Shared file and printer access over the network.
1524	TCP	Bind Shell	Metasploitable Root Shell	Open	Gives a remote shell (built in on purpose for practice).
2049	TCP	NFS	NFS v2-v4	Open	Remote file system sharing.
2121	TCP	FTP	ProFTPD 1.3.1	Open	A second FTP service for file transfer.
3306	TCP	MySQL	MySQL 5.0.51a	Open	Stores application data.

Task 3 – Vulnerability Research

Here are the most serious vulnerabilities I found while researching the services above.

Service	CVE ID	Vulnerability	CVSS	Severity	Description	Fix
vsftpd 2.3.4 (Port 21)	CVE-2011-2523	VSFTPD Backdoor Command Execution	10.0	Critical	A backdoor was secretly added to the VSFTPD 2.3.4 source code. An attacker can get remote command execution using a specially crafted username.	Remove this version right away and upgrade to a clean, supported version of VSFTPD.
UnrealIRCd (Port 6667)	CVE-2010-2075	UnrealIRCd Backdoor Command Execution	10.0	Critical	A trojanized copy of UnrealIRCd contains a hidden backdoor that lets an attacker run system commands remotely.	Install a clean, official version of UnrealIRCd and check the file integrity before installing.
Samba 3.0.20 (Port 445)	CVE-2007-2447	Samba "Username Map Script" Command Execution	10.0	Critical	A flaw in Samba lets an attacker run arbitrary commands by sending specially crafted usernames during authentication.	Upgrade Samba to a patched version and turn off the username mapping feature if it isn't needed.

Task 4 – Comparing Nmap Scan Types

I compared the different Nmap scans I used earlier to see what each one is best for.

Scan	Speed	Info Collected	Advantages	Disadvantages
nmap 10.0.2.4	Fast	Open ports	Very fast and simple.	No service or OS information.
nmap -sV 10.0.2.4	Fast	Open ports and service versions	Good starting point for vulnerability research.	A bit slower than a basic scan.
nmap -O 10.0.2.4	Medium	Operating system	Identifies the target's OS.	May not always be accurate.
nmap -A 10.0.2.4	Slow	OS, services, scripts, traceroute	Collects the most information in one go.	Slow and generates more network traffic.
nmap -p- 10.0.2.4	Slow	All TCP ports	Finds services running on every port.	Takes the longest time to finish.

Task 5 – Security Analysis & Fixes

This table brings everything together: the risk of each open service, how it could be attacked, and how to fix it.

Port / Service	Risk	Attack Scenario	Level	Fix	How to Verify	Reference
21 / FTP (vsftpd 2.3.4)	Anonymous login and an outdated FTP server may allow unauthorized access or remote code execution.	An attacker logs in anonymously, or exploits the VSFTPD backdoor to get access.	Critical	Upgrade VSFTPD. Turn off anonymous login. Only allow FTP for trusted users, or switch to SFTP.	Re-run nmap -sV --script ftp-anon and confirm anonymous login is off and the version is updated.	CIS, NIST
22 / SSH	Older SSH versions can contain security weaknesses.	An attacker tries password guessing or exploits an old SSH vulnerability.	Medium	Update OpenSSH. Use strong passwords or SSH keys. Turn off root login.	Run nmap -sV again and confirm the OpenSSH version is updated.	OpenSSH
23 / Telnet	Telnet sends usernames and passwords as plain text.	An attacker on the same network captures login details by sniffing packets.	High	Turn off Telnet. Use SSH instead. Block port 23 with a firewall.	Verify port 23 is closed with Nmap.	NIST
80 / Apache HTTP	Outdated Apache may contain known vulnerabilities.	An attacker exploits an old Apache vulnerability to access the web server.	High	Update Apache. Remove unused modules. Keep security patches installed.	Scan again with nmap -sV and confirm the new Apache version.	Apache Foundation
139 & 445 / Samba	Old Samba version is vulnerable to remote command execution. SMB signing is also disabled.	An attacker exploits the Samba vulnerability to run commands remotely.	Critical	Upgrade Samba. Turn on SMB signing. Restrict SMB access with a firewall.	Run nmap --script smb-security-mode,smb-protocols and confirm SMB is set up correctly.	Samba Team
3306 / MySQL	Database service may be reachable by unauthorized users.	An attacker tries weak passwords or exploits an outdated MySQL version.	High	Update MySQL. Use strong passwords. Restrict remote access.	Verify MySQL is updated and only trusted hosts can connect.	Oracle
5432 / PostgreSQL	Outdated database server may expose sensitive data.	An attacker exploits a database vulnerability or	High	Update PostgreSQL. Limit network access. Enforce strong authentication.	Re-scan with Nmap and confirm only authorized systems can access it.	PostgreSQL Group

Port / Service	Risk	Attack Scenario	Level	Fix	How to Verify	Reference
		weak credentials.				
5900 / VNC	VNC traffic is weakly protected and may expose remote desktop access.	An attacker guesses the VNC password and gains remote desktop access.	High	Turn off VNC if it isn't needed. Use strong passwords. Restrict access with a firewall or VPN.	Verify port 5900 is closed or protected after making changes.	NIST
6667 / UnrealIRCd	This version contains a known backdoor.	An attacker sends a malicious command and gets remote command execution.	Critical	Remove the compromised version. Install a trusted release. Verify software integrity before deploying.	Run nmap -sV and confirm the vulnerable version is no longer installed.	UnrealIRCd
8180 / Apache Tomcat	Outdated Tomcat may contain publicly known vulnerabilities.	An attacker exploits an old Tomcat vulnerability or weak management credentials.	High	Update Tomcat. Remove default accounts. Restrict access to the management interface.	Scan again with nmap -sV and confirm the updated version and secure setup.	Apache Foundation

Task 6 – Vulnerability Assessment Report

1. Introduction

This vulnerability assessment was carried out on a Metasploitable 2 virtual machine inside a controlled lab. The goal was to find open ports, list the services running on the machine, look up known vulnerabilities for those services, and judge how risky each one is. No exploitation was carried out outside of this lab.

2. Scope

- Machine: Metasploitable 2
- Operating System: Linux (Ubuntu-based)
- Target IP Address: 10.0.2.4
- Assessment Type: Vulnerability Assessment
- Environment: VirtualBox Lab

3. Tools Used

Tool	Purpose
Nmap	Port scanning, service detection, OS detection, and vulnerability enumeration.
Kali Linux	The platform I ran the assessment from.
Metasploitable 2	The target machine.
Google	Researching CVEs and vulnerabilities.

4. Enumeration Results

System Information

Item	Value
Target IP	10.0.2.4
Operating System	Linux 2.6.x
Hostname	metasploitable.localdomain

Open Ports and Services

Port	Service	Version
21	FTP	VSFTPD 2.3.4
22	SSH	OpenSSH 4.7p1
23	Telnet	Linux Telnetd
25	SMTP	Postfix
53	DNS	ISC BIND 9.4.2
80	HTTP	Apache 2.2.8
139	NetBIOS	Samba

Port	Service	Version
445	SMB	Samba 3.0.20
3306	MySQL	5.0.51a
5432	PostgreSQL	8.3.x
5900	VNC	Protocol 3.3
6667	IRC	UnrealIRCd
8180	HTTP	Apache Tomcat 5.5

Nmap Aggressive Scan

```
nmap -A -T5 10.0.2.4
```

5. Vulnerability Findings

Several outdated and vulnerable services were found running on the target machine. The most critical findings were:

- VSFTPD 2.3.4 Backdoor
- Samba 3.0.20 Remote Command Execution
- UnrealIRCd Backdoor
- Telnet Service Enabled
- Outdated Apache Web Server

Finding 1 – FTP (Port 21)

Risk Level: Critical

VSFTPD 2.3.4 has a well-known backdoor that can allow remote command execution.

```
nmap --script ftp-anon -p21 10.0.2.4
```

Finding 2 – SSH (Port 22)

Risk Level: Medium

The OpenSSH version is outdated and should be upgraded to lower the security risk.

```
nmap -sV -p22 10.0.2.4
```

Finding 3 – Telnet (Port 23)

Risk Level: High

Telnet sends usernames and passwords as plain text, which makes it insecure.

```
nmap -sV -p23 10.0.2.4
```

Finding 4 – Samba (Ports 139 & 445)

Risk Level: Critical

Samba 3.0.20 is affected by publicly known vulnerabilities, and SMB message signing is disabled.

Finding 5 – Apache Web Server (Port 80)

Risk Level: High

Apache 2.2.8 is outdated and may contain known security vulnerabilities.

```
nmap -sV -p80 10.0.2.4
```

Finding 6 – MySQL (Port 3306)

Risk Level: High

The MySQL server is outdated and should be updated to a supported version.

```
nmap -sV -p3306 10.0.2.4
```

Finding 7 – PostgreSQL (Port 5432)

Risk Level: High

The PostgreSQL service is running an outdated version.

```
nmap -sV -p5432 10.0.2.4
```

Finding 8 – Apache Tomcat (Port 8180)

Risk Level: High

Apache Tomcat 5.5 is outdated and should be upgraded.

```
nmap -sV -p8180 10.0.2.4
```

6. Recommendations

- Update all outdated services to supported versions.
- Disable services that aren't needed, such as Telnet.
- Replace FTP with SFTP for secure file transfers.
- Enable SMB message signing to protect SMB traffic.
- Restrict access to admin services using firewall rules.
- Use strong passwords and multi-factor authentication where possible.
- Apply security patches regularly.
- Run vulnerability assessments periodically to catch new issues.

7. Evidences

No.	What It Shows	Command Used
Screenshot 1	Nmap aggressive scan result	<code>nmap -A -T5 10.0.2.4</code>
Screenshot 2	FTP enumeration result (Finding 1)	<code>nmap --script ftp-anon -p21 10.0.2.4</code>
Screenshot 3	SSH service version scan (Finding 2)	<code>nmap -sV -p22 10.0.2.4</code>
Screenshot 4	Telnet enumeration result (Finding 3)	<code>nmap -sV -p23 10.0.2.4</code>
Screenshot 5	HTTP service detection (Finding 5)	<code>nmap -sV -p80 10.0.2.4</code>
Screenshot 6	MySQL enumeration result (Finding 6)	<code>nmap -sV -p3306 10.0.2.4</code>
Screenshot 7	PostgreSQL enumeration result (Finding 7)	<code>nmap -sV -p5432 10.0.2.4</code>
Screenshot 8	Tomcat enumeration result (Finding 8)	<code>nmap -sV -p8180 10.0.2.4</code>

Screenshot 1:

```

(kali㉿kali)-[~]
$ nmap -A 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 02:18 EDT
Nmap scan report for 10.0.2.4
Host is up (0.00088s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 10.0.2.3
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)

```

Screenshot 2:

```

(kali㉿kali)-[~]
$ nmap --script ftp-anon -p21 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-01 01:19 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0011s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
MAC Address: 08:00:27:88:8D:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

```

Screenshot 3:

```

(kali㉿kali)-[~]
$ nmap -sV -p22 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-01 01:29 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0013s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
MAC Address: 08:00:27:88:8D:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

```

Screenshot 4:

```

(kali㉿kali)-[~]
$ nmap -sV -p23 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-01 01:30 EDT
Nmap scan report for 10.0.2.4
Host is up (0.00061s latency).

PORT      STATE SERVICE VERSION
23/tcp    open  telnet   Linux telnetd
MAC Address: 08:00:27:88:8D:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

```

Screenshot 5:

```
(kali㉿kali)-[~]
$ nmap -sV -p80 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-01 01:32 EDT
Nmap scan report for 10.0.2.4
Host is up (0.00062s latency).

PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.2.8 ((Ubuntu) DAV/2)
MAC Address: 08:00:27:88:8D:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

Screenshot 6:

```
(kali㉿kali)-[~]
$ nmap -sV -p3306 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-01 01:33 EDT
Nmap scan report for 10.0.2.4
Host is up (0.00047s latency).

PORT      STATE SERVICE VERSION
3306/tcp  open  mysql     MySQL 5.0.51a-3ubuntu5
MAC Address: 08:00:27:88:8D:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

Screenshot 7:

```
(kali㉿kali)-[~]
$ nmap -sV -p5432 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-01 01:35 EDT
Nmap scan report for 10.0.2.4
Host is up (0.00058s latency).

PORT      STATE SERVICE      VERSION
5432/tcp  open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
MAC Address: 08:00:27:88:8D:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

Screenshot 8:

```
(kali㉿kali)-[~]
$ nmap -sV -p8180 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-01 01:36 EDT
Nmap scan report for 10.0.2.4
Host is up (0.00046s latency).

PORT      STATE SERVICE      VERSION
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:88:8D:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

8. Conclusion

This assessment found several outdated services and insecure settings on the target system. Multiple Critical, High, and Medium severity issues were found, including vulnerable FTP, Samba, and UnrealIRCd services. If left unpatched, these issues could lead to unauthorized access, remote command execution, or information disclosure.

Applying the recommended fixes – updating software, removing unnecessary services, and following good security practices – will reduce the attack surface and improve the overall security of the system.

Overall Risk Rating: Critical